Spectral Tilt Without Specificity: Placement Versus Capacity in the Dual Geometry of Large Language Models

Nuraj Rimal

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Abstract

A recent account of transformer geometry holds that concepts "whisper" and syntax "shouts": concept directions concentrate in the low-variance tail of the representation spectrum, and syntactic information in the high-variance head (Acharya et al., 2026). We test both halves with the same instrument. For each tail we ask two questions. The first is about *placement*: does the measured quantity sit where the account says it does? Under the published measurement it does in both tails, though the tilt is small and depends on the basis and the null. The second is about *capacity* and *specificity*: is the label better recovered from that tail than from the other one, and more so than a matched control label that carries none of the claimed content? We test this across eight open-weight models, eleven behavioural concepts and six window sizes. Part of speech prefers the head no more than a random lexical partition does (mean $\Delta = -0.040$, 95% CI $[-0.073, -0.007]$). Concepts prefer the tail no more than surface length does (mean $\Delta = -0.003$, TOST $p = 0.005$), and the effect lies inside a pre-registered equivalence bound of ± 0.10 retention units. Every result is reported with and without the model that failed our reliability gate, at six window sizes, at two layers, under length-matched and length-residualised controls, and with a non-linear probe. The window size, the primary tests, the equivalence bound and the multiplicity correction were frozen before the new data were collected (commits `2bcfd96` and `da3de55`). We also report the methodological problems we hit along the way: cross-validation leakage between near-duplicate statement pairs, retention denominators that turn noise into effects, and isotropic baselines that overstate spectral effects. Each one, left unfixed, would have produced a confident but wrong result.

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Chapter 1

Introduction

The linear representation hypothesis holds that language models encode high-level concepts as directions in activation space (Park et al., 2024; Mikolov et al., 2013). A sharper geometric claim has followed. Acharya et al. (2026) report that concept directions *anti-concentrate*: their energy sits in the low-variance tail of the unembedding second-moment matrix rather than in the high-variance head. A part-of-speech probe, meanwhile, reads syntax more accurately from the head than from the tail. The resulting picture, "concepts whisper while syntax shouts", is a claim about where in the spectrum different kinds of information live.

The claim has two halves, and each half has two parts. It says where a quantity is *placed* (the tail for concepts, the head for syntax), and it implies that this placement is *specific*: that the tail holds concepts *as concepts* and the head holds syntax *as syntax*, rather than either tail simply being where information of any kind ends up. The published evidence addresses placement. Specificity requires a control: a label that is equally easy to read out of the representation but carries none of the claimed content. This is the problem that control tasks were introduced to solve for probing in general (Hewitt and Liang, 2019). A probe that reads a property from a representation shows only that the information is there, not that the representation is organised around it (Belinkov, 2022; Ravichander et al., 2021; Elazar et al., 2021).

This thesis brings that standard to both halves of the spectral claim, using the same estimator, folds, normalisation and bootstrap on each side so that the two halves are measured on one scale.

Main result. *Under the published measurement both tails tilt; neither tilt is specific.* The published picture reproduces when it is measured the published way: concept directions sit in the tail against an isotropic baseline, and part of speech is read more accurately from the head in 6 of 8 models. Both tilts are fragile. The syntactic tilt is not significant and reverses in the full-vocabulary basis, and against structural nulls concept directions sit in the tail in only about half the cells. Specificity does not hold at all. Against matched controls, the head prefers syntax no more than it prefers an arbitrary partition of the vocabulary, and the tail prefers concepts no more than it prefers surface length.

Contributions.

1. A symmetric two-tail test of spectral specificity, a difference in differences of *retention* across the spectrum, with a matched control for each tail (Chapter 3).

2. Pre-registered evidence across eight models and eleven concepts that neither tail is specific, including an equivalence test that turns "we found nothing" into "the effect is smaller than ± 0.10" (Chapter 5).

3. A direct comparison of spectral *placement* (where a direction's energy sits) with subspace *capacity* (where the concept can be read from), which the original account treated as one quantity (Section 5.3).

4. Methodological findings that apply beyond this paper, each of which on its own would have produced a confident but wrong result (Chapter 6).

Chapter 2

Related Work

Linear representations and spectral geometry. Park et al. (2024) formalise the linear representation hypothesis and the role of the unembedding geometry; Acharya et al. (2026) add the dual, spectral reading that this thesis tests. Anisotropy in contextual representations (Ethayarajh, 2019) and a small number of rogue, high-variance dimensions (Timkey and van Schijndel, 2021) mean that the high-variance head is dominated by a few directions, which is why the choice of null baseline matters so much for any spectral statistic.

Probing and its controls. A probe's accuracy shows that a property is decodable, not that the model uses it or is organised around it (Belinkov, 2022; Ravichander et al., 2021). Hewitt and Liang (2019) introduced *control tasks*: labels assigned to word types at random, so they can be learned only by memorising lexical identity. They proposed *selectivity*, the accuracy gap between the real task and its control, as the quantity to report. Our loud-tail control, `token_id_random`, is a control task in exactly this sense, with class sizes matched to the part-of-speech distribution. Information-theoretic (Pimentel et al., 2020) and description-length (Voita and Titov, 2020) probing make the same point from a different angle: probe accuracy has to be read against what the probe could have learned anyway. Conditional probing (Hewitt et al., 2021) asks what a representation adds beyond a baseline. Our difference in differences asks the analogous question across spectral subspaces.

Removing information. Amnesic probing (Elazar et al., 2021), iterative nullspace projection (Ravfogel et al., 2020) and closed-form concept erasure (Belrose et al., 2023) test whether a property *matters* by removing it. Our length-residualised control (Section 5.4) uses the same move on a nuisance variable: it removes the linear length signal from the activations and asks whether the concept result changes.

Steering. Activation addition (Turner et al., 2023; Panickssery et al., 2024) and the refusal direction (Arditi et al., 2024) are the behavioural counterparts of these geometric claims. Our steering experiments are reported in Appendix C.

Chapter 3

Method

3.1 Spectral basis

Let $\mathbf{W}_U \in \mathbb{R}^{|\mathcal{V}|\times d}$ be the unembedding matrix. The reference matrix is the second moment $\mathbf{\Sigma} = \frac{1}{|\mathcal{L}|}\sum_{\ell\in\mathcal{L}} \boldsymbol{\gamma}_\ell \boldsymbol{\gamma}_\ell^\top$ over a token set $\mathcal{L} \subseteq \mathcal{V}$, where $\boldsymbol{\gamma}_\ell \in \mathbb{R}^d$ is the unembedding vector of token ℓ (the ℓ-th row of $\mathbf{W}_U$), with eigendecomposition $\mathbf{\Sigma} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$, $\lambda_1 \geq \cdots \geq \lambda_d$. The primary analyses use the published language-subset basis, so that both tails are measured in the basis in which the published claims were made. The full-vocabulary basis ($\mathcal{L} = \mathcal{V}$) is a pre-specified robustness check.

3.2 Placement

For a unit concept direction $\mathbf{v}$ the energy CDF is $C(k) = \sum_{i\leq k}(\mathbf{v}^\top \mathbf{u}_i)^2$ and the Gini deviation is $\mathcal{G} = \int_0^1 [C(V) - V]\, dV$, where $V(k)$ is the cumulative explained variance. Negative $\mathcal{G}$ means energy in the tail. We report $\mathcal{G}$ against two nulls: norm-matched isotropic Gaussian directions, as published, and structural nulls built from the same data, namely per-pair sign flips for matched contrasts and label repartitions for unmatched ones (Appendix A).

3.3 Capacity and specificity

Placement describes one vector. Capacity describes a subspace: how much of a label can be read from it. For a window of k consecutive eigenvectors starting at position s, $\mathbf{U}_{s:s+k}$, we train a probe on $\mathbf{X}\mathbf{U}_{s:s+k}$ and report *retention*,

$$R_y(s) = \frac{\mathrm{acc}_y(s) - m_y}{\mathrm{acc}_y(\text{full } d) - m_y}, \tag{3.1}$$

where m_y is the majority-class rate. Retention is the share of the label's own full-dimensional signal that survives the projection. It puts labels with different ceilings on one scale. Two headroom thresholds apply, and they do different jobs. A retention value is *computed* only when the full-dimensional probe beats the majority rate by at least 0.02; below that the denominator is noise and the ratio is undefined. Separately, a quiet-tail cell *enters the aggregate* only if the concept's headroom is at least 0.10 and the shuffled-label floor check passes (Section 3.4). The

first is a numerical guard; the second, pre-registered, decides which computed values are reliable enough to average.

The *tilt* of a label toward a tail is the retention gap between the two ends of the spectrum. For the loud tail $g_y = R_y(\text{top}) - R_y(\text{bottom})$; for the quiet tail the sign is flipped. *Specificity* is the difference in differences between the real label and its control,

$$\Delta = g_{\text{real}} - g_{\text{control}}, \tag{3.2}$$

with a 95% interval from a paired bootstrap over items (10,000 resamples), so that one resample is applied to all labels at once. $\Delta > 0$ means the tail prefers the real label more than it prefers the control.

Loud-tail control. `token_id_random` assigns every word *type* to one of C classes at random, fixed across its occurrences, with class sizes matched to the part-of-speech distribution by bin packing. It contains no syntax but is exactly as recoverable from lexical identity as part of speech is. Type frequency and word length are reported as second and third controls; both correlate with part of speech, so they bound the confound rather than remove it.

Quiet-tail control. Folds are grouped by statement pair (Section 6.2), so both members of a pair are always held out together. This rules out the direct analogue of the loud-tail control. A random partition of *pairs* can only be learned by recognising a held-out pair's mate in the training fold, which grouping prevents, so it sits at chance; we keep it as a leakage detector, not a control. A label that is constant within a pair is still learnable when it reflects a general property of the statements rather than pair identity: the probe can learn it from other pairs and apply it to unseen ones. Surface length is such a property. In the sycophancy sets both members of a pair have identical length (Table 5.3), yet under grouped folds length is read from the activations of the CAA sycophancy set with 0.97 accuracy in every model. We therefore use a median split on character length as the control: it is learnable, carries no concept content by construction, and is available for every dataset. Its weakness is that it can correlate with the label within a pair; Section 5.4 measures that correlation and removes it.

Probe. Multinomial logistic regression (L-BFGS, $C = 1$, standardised inputs, five folds), identical to the published part-of-speech probe. As a robustness check, a one-hidden-layer MLP (256 units) is run on the same windows and folds.

3.4 Inference and pre-registration

Models are the unit of analysis. In the quiet tail, concepts are averaged within a model before testing, because concepts in one model share an eigenbasis and are not independent. We use one-sample t-tests over models and report Wilcoxon signed-rank tests alongside.

The following were frozen in `prereg/two_tail_prereg.json` and committed (`da3de55`) before any of the new concepts, window sizes or layers were run.[1] They extend, and do not change, the original pre-specification (`2bcfd96`).

[1]The public repository is a single release snapshot. The full development history, including commits `2bcfd96` and `da3de55`, is archived and available on request; `prereg/two_tail_prereg.lock.json` records the SHA-256 and timestamp of the freeze.

- **Window size.** $k = 0.10\,d$ for the loud tail (the published setting) and $k = 0.005\,d$ for the quiet tail (the setting of the original concept runs). The sweep $k \in \{0.25, 0.5, 1, 2, 5, 10\}\%$ is reported in full as robustness; no conclusion is drawn from any other k.
- **One primary test per tail.** Loud: Δ against `token_id_random`, final layer. Quiet: Δ against length, mid-depth layer.
- **Multiplicity.** Two tails $\times$ two layers $\times$ two quantities (tilt g and specificity Δ) form a family of eight, corrected with Holm–Bonferroni (Holm, 1979). A member that was not run enters with $p = 1$.
- **Equivalence.** TOST (Lakens, 2017) with a smallest effect size of interest of ± 0.10 retention units, the smallest difference we would describe as a tail "holding" a label. It is distinct from the ± 0.082 Gini bound used for placement, which is on a different scale. Because specificity is directional, we also report the one-sided non-superiority test of $H_0 : \Delta \geq +0.10$.
- **gemma-2-2b.** The model failed the split-half reliability gate on the concept side (Appendix B). Every aggregate is reported with and without it, and a conclusion is called robust only if it holds both ways.
- **Cell exclusion.** A quiet-tail cell enters the aggregate only if its shuffled-label floor check passes and the concept's full-dimensional headroom is at least 0.10. Table 5.1 gives the number of usable cells per model.

Chapter 4

Experimental Setup

Models. Qwen2.5-0.5B, Qwen2.5-3B, Qwen2.5-7B, Qwen3-4B, Mistral-7B-v0.3, Llama-3.2-3B, Llama-3.1-8B and gemma-2-2b. PyTorch `2.11.0+cu128`, Transformers `5.18.0`.

Loud tail. Universal Dependencies English-EWT (Silveira et al., 2014; Nivre et al., 2020), first 500 sentences, final-layer token activations with the final norm applied (sub-word mean pooling, published tag filter, tags with ≥ 30 occurrences). A mid-depth read (hidden state $\lfloor L/2 \rfloor$, no final norm) is the second layer. A reproduction gate requires the top-k and bottom-k accuracies to match the published values within 0.02 before any control is read; in practice they match far more closely (Section 5.1).

Quiet tail. Eleven matched-pair concepts, up to 512 pairs each, last-token pooling, mid-depth and final layers: the truth datasets of Marks and Tegmark (2024) (*cities*, *cities_negated*, *conjunction*, *disjunction*, *counterfact*, *magnitude*, *translation*) and the sycophancy sets of Panickssery et al. (2024) and Perez et al. (2022) (*sycophancy* and its *nlp*, *philpapers* and *political* subsets). Refusal is unmatched (harmful and harmless instructions come from different corpora), so it has no within-pair control and enters only the placement analysis.

Source	Concepts	Pairs used	Matched
`geometry-of-truth` (Marks and Tegmark, 2024)	7 truth sets	≤ 512 each	yes
`CAA` (Panickssery et al., 2024)	sycophancy	512	yes
`evals` (Perez et al., 2022)	3 sycophancy subsets	500 each	yes
`refusal_direction` (Arditi et al., 2024)	4 refusal sets	100–256	no
UD English-EWT	part of speech	6,585 tokens	–

Table 4.1: Data. Refusal enters the placement analysis only.

Chapter 5

Results

5.1 Under the published measurement, both tails tilt

Placement reproduces. Following the published isotropic method, the mean word-level Gini deviation is -0.397, against a published -0.356 (Acharya et al., 2026). The unembedding second moment has condition number 2.30×10^3 and an isotropic random-direction baseline of -0.1452. Our extraction matches the published routine at cosine 1.000 on Qwen2.5-0.5B.

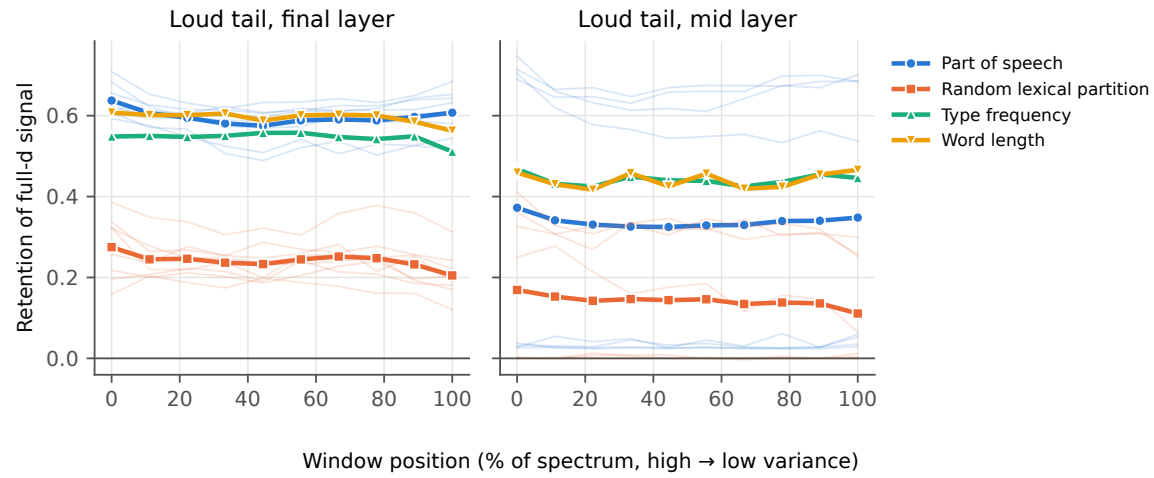
Capacity reproduces. The part-of-speech probe reproduces the published one almost exactly. On the 3 models whose per-model accuracies were published (Llama-3.2-3B, Qwen2.5-3B, gemma-2-2b), our top-k and bottom-k accuracies differ from the published values by at most 0.0005. Before any new result was read, a smoke run reproduced the thesis's own earlier numbers (Qwen2.5-0.5B: loud $\Delta = +0.0264$ and quiet $\Delta = +0.1704$ / $+0.0844$, matched to four decimals). The null results that follow therefore come from the published pipeline, not from a reimplementation that could have lost the effect.

The loud tail tilts. Part of speech is read more accurately from the top-k window than from the bottom-k window in 6 of 8 models, as published. The mean retention gap is $g = +0.030$ (95% CI $[-0.005, +0.065]$, $p = 0.079$). The tilt is basis-dependent: in the full-vocabulary basis the mean gap is -0.019 ($p = 0.333$), with part of speech read better from the *bottom* of the spectrum in several models.

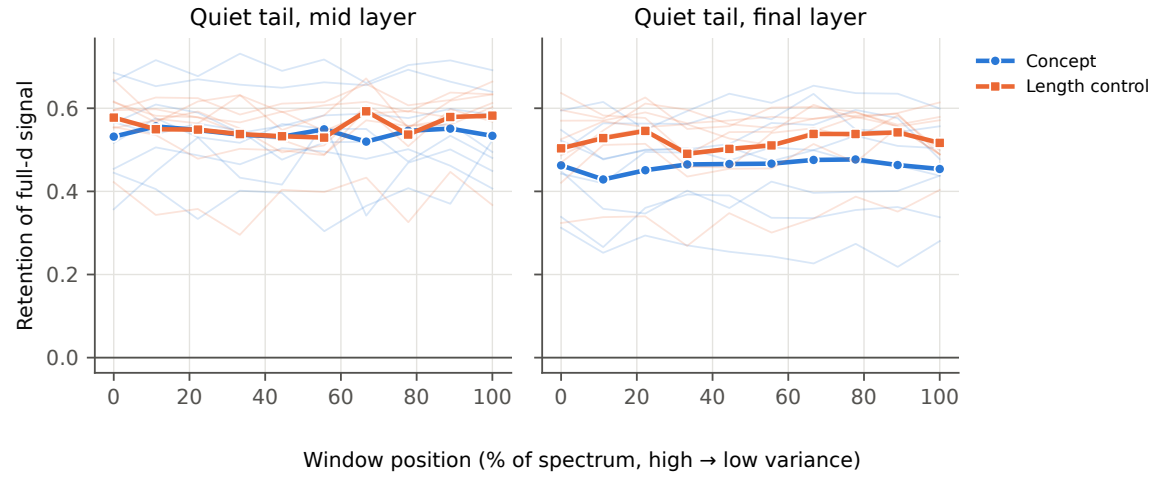
The quiet tail tilts in placement. Concept directions carry more of their energy in the tail than isotropic directions do. Against structural nulls, however, the effect shrinks to a few hundredths of a Gini unit, and the direction sits in the tail in only 30 of 60 (model, concept) cells (Section 5.3, Appendix A).

5.2 Neither tilt is specific

Figure 5.1 shows retention across the whole spectrum. In both tails the real label and its control trace curves of nearly the same shape: the gap between them does not open at the end the account predicts.



Window size k = 0.1·d. Thick: mean over models; thin: individual models (POS, random partition).



Window size k = 0.005·d. Thick: mean over models (concepts averaged within model); thin: individual models.

Figure 5.1: Retention of each label's full-dimensional signal in a k-dimensional window, swept from the high-variance head (left) to the low-variance tail (right). Top: part of speech and three controls. Bottom: concepts and the length control. Thick lines are means over models; thin lines are individual models. The published comparison is the first window against the last.

Primary tests. *Loud tail.* Part of speech prefers the head no more than a random lexical partition does: $\Delta = -0.040$ (95% CI $[-0.073, -0.007]$, $p = 0.025$, $n = 8$); without gemma-2-2b, $\Delta = -0.034$ ($p = 0.062$). The control is preferred more than part of speech in the all-model version only, so we claim no more than that part of speech is not preferred. The one-sided test rejects any syntax-specific advantage as large as $+0.10$ (non-superiority $p < 0.001$; without gemma $p < 0.001$).

Quiet tail. Concepts prefer the tail no more than surface length does: $\Delta = -0.003$ (95% CI $[-0.068, +0.062]$, $p = 0.920$, $n = 8$ models); without gemma-2-2b, $\Delta = -0.009$ ($p = 0.774$). The 90% interval is $[-0.055, +0.049]$ (TOST $p = 0.005$; without gemma $p = 0.013$). This interval lies inside the pre-registered bound of ± 0.10 in both versions, so the quiet-tail specificity effect is not just undetected but smaller than the smallest effect we said would matter.

Multiplicity. Table 5.2 lists all eight members of the pre-registered family. After Holm correction, with all eight models, no member is significant; without gemma-2-2b, no member is significant.

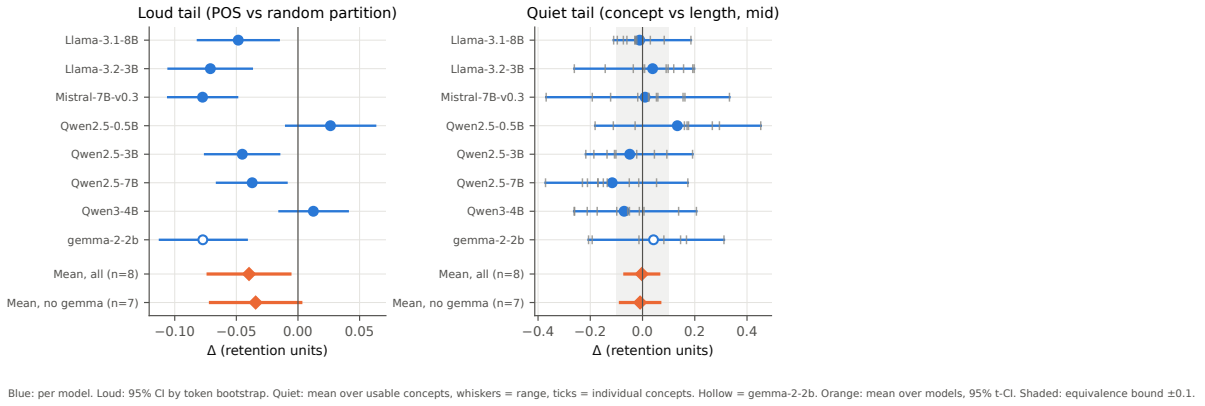


Figure 5.2: Specificity Δ per model. Left: loud tail, part of speech against a random lexical partition, 95% CI by token bootstrap. Right: quiet tail, concept against length, mean over concepts with the range across concepts. Hollow: gemma-2-2b. Orange: mean over models with a 95% t-interval, with and without gemma-2-2b. Shaded: the ± 0.10 equivalence bound.

5.3 Placement and capacity

Placement and capacity answer different questions. A difference-of-means direction records where along the spectrum the class *means* differ; a probe on a window records whether the classes are *separable* there, which also depends on how much within-class variance that window carries. Figure 5.3 sets the two side by side for every (model, concept) cell measured both ways (60 cells; Spearman $\rho = -0.39$, descriptive only, since cells within a model share a basis). Placement and capacity are moderately related, in the expected direction: cells whose direction sits further in the tail tend to read the concept better from the tail. The relation is far from one-to-one: against the structural null the direction's energy sits in the tail in 30 of 60 cells, and in 15 of those is the concept also read better from the tail than from the head. And where

Model	Loud Δ	95% CI	Quiet Δ (mid)	concepts
Qwen2.5-0.5B	+0.026	[-0.010, +0.063]	+0.133	9/11
Qwen2.5-3B	-0.045	[-0.075, -0.015]	-0.049	9/11
Qwen2.5-7B	-0.037	[-0.066, -0.009]	-0.116	11/11
Qwen3-4B	+0.012	[-0.015, +0.040]	-0.070	11/11
Mistral-7B-v0.3	-0.077	[-0.105, -0.049]	+0.010	11/11
Llama-3.2-3B	-0.071	[-0.105, -0.037]	+0.039	11/11
Llama-3.1-8B	-0.048	[-0.081, -0.016]	-0.011	11/11
gemma-2-2b†	-0.077	[-0.112, -0.042]	+0.042	7/11

Table 5.1: Per-model specificity Δ for both tails at the pre-registered primary settings. Loud: part of speech against a random lexical partition, final layer, $k = 0.1\,d$, 95% CI by token bootstrap. Quiet: concept against the length control, mid layer, $k = 0.005\,d$, mean over usable concepts (usable/run). †gemma-2-2b failed the reliability gate on the concept side; every aggregate is reported with and without it.

Tail	Layer	Quantity	mean	95% CI	p	p_{Holm}	p_{TOST}
Loud	final	Δ (primary)	−0.040	[−0.073, −0.007]	0.025	0.203	0.002
Loud	final	g	+0.030	[−0.005, +0.065]	0.079	0.476	0.001
Loud	mid	Δ	−0.034	[−0.063, −0.004]	0.030	0.209	<0.001
Loud	mid	g	+0.024	[−0.029, +0.077]	0.314	1.000	0.006
Quiet	mid	Δ (primary)	−0.003	[−0.068, +0.062]	0.920	1.000	0.005
Quiet	mid	g	+0.002	[−0.050, +0.053]	0.932	1.000	0.001
Quiet	final	Δ	−0.022	[−0.069, +0.026]	0.314	1.000	0.003
Quiet	final	g	−0.009	[−0.040, +0.022]	0.531	1.000	<0.001

Table 5.2: The pre-registered family, all eight models. The same table without gemma-2-2b is Table B.1 (Appendix B). g: tilt of the real label; Δ: specificity. TOST bound ±0.10.

capacity does favour the tail, it favours the length control just as much, so placement predicts where information is recoverable, not where the concept specifically lives.

5.4 Robustness

Window size. Figure 5.4 reports Δ at every window size from 0.25% to 10% of d, for both tails and both layers. Individual cells change sign with k, which is why k was fixed in advance. The mean Δ is significantly positive in 1 of 48 tail × layer × window × model-set cells, against about 1.2 expected by chance alone at two-sided $\alpha = 0.05$: loud mid at $k = 0.5\%$ (all models only). It is significantly *negative* (the control preferred over the real label) in 4 cells, at 3 settings: loud final at $k = 5\%$ (all models only), loud final at $k = 10\%$ (all models only) and loud mid at $k = 10\%$ (with and without gemma-2-2b). No cell is interpreted on its own, because k was fixed in advance; at the pre-registered k the conclusion is as stated above, and no window size supports a specific effect.

Controls. Part of speech also fails to beat type frequency and word length: type frequency ($\Delta = -0.007$, $p = 0.609$; it beats the control in 3 of 8 models); word length ($\Delta = -0.015$, $p = 0.331$; 3 of 8).

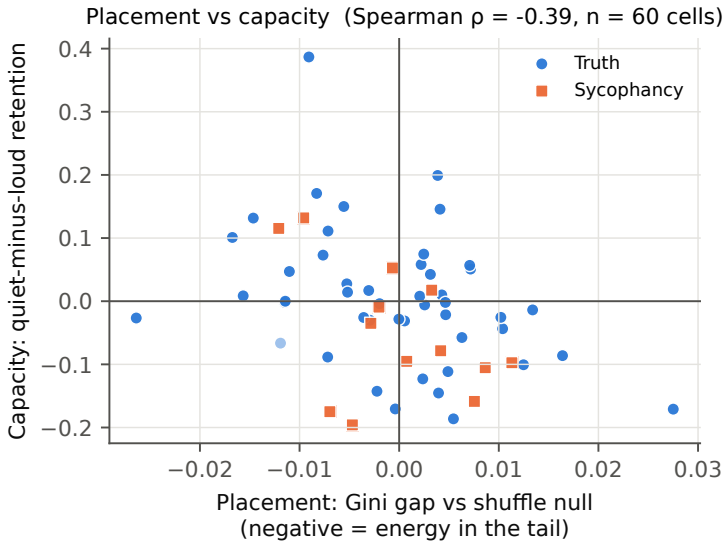


Figure 5.3: Placement (Gini gap against the structural shuffle null; negative means energy in the tail) against capacity (quiet-minus-loud retention), one point per (model, concept) cell at the primary setting.

Length. Table 5.3 reports how strongly length alone predicts each concept label. Restricting to a subset of pairs in which the positive member is the longer one exactly as often as the negative gives $\Delta = +0.013$ (TOST $p = 0.027$). Regressing length out of the activations the concept probe sees gives $\Delta = -0.007$ (TOST $p = 0.004$). Both stay inside the ± 0.10 equivalence bound, so the length control's correlation with the label does not drive the quiet-tail result.

Layer. At the other layer the specificity effect is $\Delta = -0.034$ (loud, mid) and $\Delta = -0.022$ (quiet, final) (Table 5.2).

Non-linear probe. With a one-hidden-layer MLP on four models (Qwen2.5-3B, Llama-3.2-3B, Mistral-7B-v0.3, Qwen3-4B), $\Delta = 0.000$ (95% CI $[-0.022, +0.022]$, TOST $p < 0.001$) in the loud tail and $\Delta = -0.034$ ($[-0.086, +0.018]$, TOST $p = 0.014$) in the quiet tail, so the conclusion is not specific to linear read-out. With $n = 4$ models these intervals are wide, and the check rules out a large non-linear effect rather than any effect.

Basis. In the full-vocabulary basis, with all eight models rerun into a folder that the analysis audits for mixed bases, the loud-tail $\Delta = -0.010$ ($p = 0.636$). This replaces the earlier full-vocabulary figure, which was computed from a folder in which two of the eight result files had been produced in the published basis.

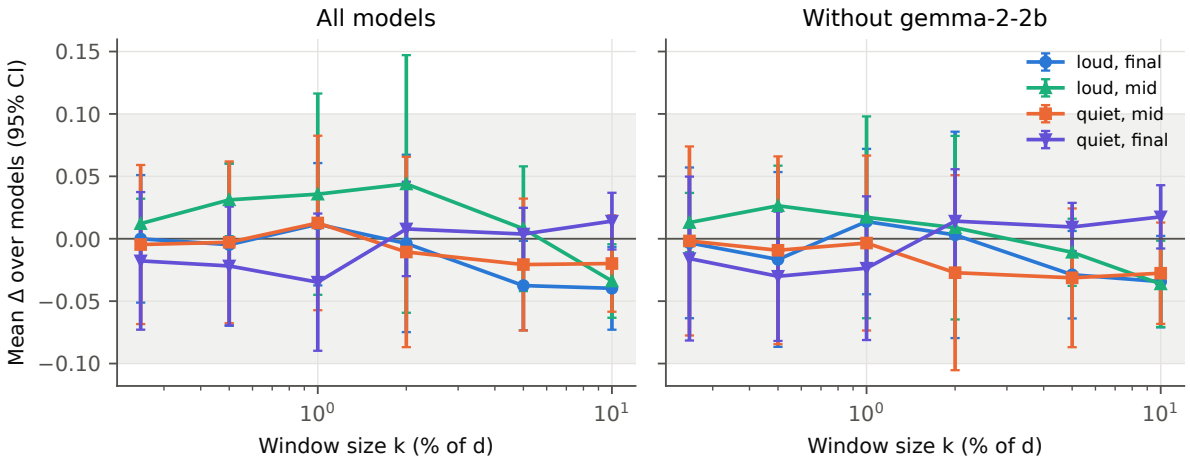


Figure 5.4: Mean Δ over models (95% CI) against window size, both tails and both layers, with all models (left) and without gemma-2-2b (right). Shaded: equivalence bound.

Dataset	pairs	AUC chars	AUC tokens	pos. longer	tied
`sycophancy`	512	0.500	0.500	0%	100%
`sycophancy_nlp`	500	0.500	0.500	0%	100%
`sycophancy_philpapers`	500	0.500	0.500	0%	100%
`sycophancy_political`	500	0.500	0.500	0%	100%
`truth_cities`	512	0.487	0.498	40%	19%
`truth_cities_negated`	512	0.513	0.502	41%	19%
`truth_conjunction`	512	0.489	0.515	46%	4%
`truth_counterfact`	512	0.495	0.500	36%	24%
`truth_disjunction`	512	0.480	0.515	44%	5%
`truth_magnitude`	512	0.640	0.500	87%	0%
`truth_translation`	354	0.504	0.506	50%	0%

Table 5.3: How much of each concept label is predictable from length alone. AUC = 0.5 means length is uninformative.

Chapter 6

Methodological Findings

Each of these, left in place, would have produced a confident but wrong result. We report them because they are not specific to this paper.

6.1 Isotropic baselines overstate spectral effects

In a strongly anisotropic spectrum, a norm-matched random direction already has a Gini deviation of -0.16 to -0.28. A direction can therefore look anti-concentrated against the isotropic null while being indistinguishable from a sign-flip or label-repartition null built from the same data (Appendix A). Both nulls should be reported; choosing the one that gives the more favourable answer is not an option.

6.2 Leakage between near-duplicate pairs

Statement pairs share most of their tokens (sycophancy pairs share 99.2%). A statement-level split puts a test item's pair-mate in the training fold with the opposite label about 80% of the time. The probe learns the mate and answers the test item confidently backwards, so a linear probe scores *below* chance (0.18 on a balanced binary task in gemma-2-2b). In the other direction, the same leakage inflates any pair-level control label. Pair-grouped folds remove both effects; a pair-level random label then sits at chance and serves as a leakage detector.

6.3 Retention denominators

Normalising by full-dimensional headroom puts labels on one scale, but when the headroom is small it turns a two-point accuracy difference into a retention gap of -0.47 (Qwen2.5-0.5B on sycophancy, headroom 0.045). Cells with concept headroom below 0.10 are excluded, as pre-registered; per-model counts of usable cells are given in Table 5.1, and the summary file `concept_subspace_summary.csv` reports every test with and without the exclusion.

6.4 Degenerate bottom eigenspaces

When regularisation $\lambda\mathbf{I}$ dominates the small eigenvalues, the bottom of the spectrum collapses onto a plateau within which the eigenvectors are arbitrary, so "the bottom-k subspace" names

no reproducible object. At $\lambda = 0.1$ the condition number fell to 1.38 and the random-direction Gini to -0.0004. All reported runs check for this; none of the primary bases is degenerate.

Chapter 7

Limitations

- **The pre-registered quiet-tail control is surface form.** We measure length's correlation with each label and test whether the result survives balancing and residualising it. A semantic nuisance label annotated on the same truth statements would be the strongest control and does not exist in these datasets.
- **Eight models.** Models are the unit of analysis, so $n = 8$ (or 7). Effects smaller than the 90% intervals reported here cannot be ruled out at this n.
- **Two layers.** We probe the final and mid-depth layers, not the full depth profile.
- **Probes measure decodability.** A null specificity result says the tails do not *hold* labels selectively; it does not say the model does not *use* spectral position. Interventional tests (Elazar et al., 2021; Belrose et al., 2023) are the natural next step.
- **One syntactic label, one language.** Part of speech on English EWT. Dependency relations and other languages are untested here.
- **gemma-2-2b** fails the concept-side reliability gate, and its word concepts never separate at any layer. We report it both ways throughout rather than resolving why.
- **Refusal** is unmatched and enters only the placement analysis.

Chapter 8

Conclusion

Measured the way it was published, the spectral account holds, if weakly: concept directions sit in the tail against an isotropic baseline, and syntax is read slightly more easily from the head in the published basis. Measured with a control on each side, using one estimator for both, neither tail is specific. The head prefers syntax no more than it prefers an arbitrary partition of the vocabulary, and the tail prefers concepts no more than it prefers length. Spectral claims about representations need the same control-task discipline that probing claims already require.

Bibliography

Pratyush Acharya, Nuraj Rimal, and Habish Dhakal. Concepts whisper while syntax shouts: Spectral anti-concentration and the dual geometry of transformer representations. *arXiv preprint arXiv:2605.01609*, 2026.

Andy Arditi, Oscar Obeso, Aaquib Syed, Daniel Paleka, Nina Rimsky, Wes Gurnee, and Neel Nanda. Refusal in language models is mediated by a single direction. In *Advances in Neural Information Processing Systems*, 2024.

Yonatan Belinkov. Probing classifiers: Promises, shortcomings, and advances. *Computational Linguistics*, 48(1):207–219, 2022.

Nora Belrose, David Schneider-Joseph, Shauli Ravfogel, Ryan Cotterell, Edward Raff, and Stella Biderman. LEACE: Perfect linear concept erasure in closed form. In *Advances in Neural Information Processing Systems*, 2023.

Yanai Elazar, Shauli Ravfogel, Alon Jacovi, and Yoav Goldberg. Amnesic probing: Behavioral explanation with amnesic counterfactuals. *Transactions of the Association for Computational Linguistics*, 9:160–175, 2021.

Joshua Engels, Eric J. Michaud, Isaac Liao, Wes Gurnee, and Max Tegmark. Not all language model features are one-dimensionally linear. In *International Conference on Learning Representations*, 2025.

Kawin Ethayarajh. How contextual are contextualized word representations? comparing the geometry of BERT, ELMo, and GPT-2 embeddings. *arXiv preprint arXiv:1909.00512*, 2019.

John Hewitt and Percy Liang. Designing and interpreting probes with control tasks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 2733–2743, 2019.

John Hewitt, Kawin Ethayarajh, Percy Liang, and Christopher D. Manning. Conditional probing: Measuring usable information beyond a baseline. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, 2021.

Sture Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70, 1979.

Daniël Lakens. Equivalence tests: A practical primer for t tests, correlations, and meta-analyses. *Social Psychological and Personality Science*, 8(4):355–362, 2017. doi: 10.1177/1948550617697177.

Samuel Marks and Max Tegmark. The geometry of truth: Emergent linear structure in large language model representations of true/false datasets. *arXiv preprint arXiv:2310.06824*, 2024.

Tomas Mikolov, Kai Chen, Greg Corrado, and Jeffrey Dean. Efficient estimation of word representations in vector space. *arXiv preprint arXiv:1301.3781*, 2013.

Joakim Nivre, Marie-Catherine de Marneffe, Filip Ginter, Jan Hajič, Christopher D. Manning, Sampo Pyysalo, Sebastian Schuster, Francis Tyers, and Daniel Zeman. Universal Dependencies v2: An evergrowing multilingual treebank collection. In *Proceedings of the Twelfth Language Resources and Evaluation Conference*, 2020.

Nina Panickssery, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Matt Turner. Steering Llama 2 via contrastive activation addition. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15504–15522, 2024.

Kiho Park, Yo Joong Choe, and Victor Veitch. The linear representation hypothesis and the geometry of large language models. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *PMLR*, 2024.

Ethan Perez, Sam Ringer, Kamilė Lukošiūtė, et al. Discovering language model behaviors with model-written evaluations. *arXiv preprint arXiv:2212.09251*, 2022.

Tiago Pimentel, Josef Valvoda, Rowan Hall Maudslay, Ran Zmigrod, Adina Williams, and Ryan Cotterell. Information-theoretic probing for linguistic structure. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 2020.

Shauli Ravfogel, Yanai Elazar, Hila Gonen, Michael Twiton, and Yoav Goldberg. Null it out: Guarding protected attributes by iterative nullspace projection. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 2020.

Abhilasha Ravichander, Yonatan Belinkov, and Eduard Hovy. Probing the probing paradigm: Does probing accuracy entail task relevance? In *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics*, 2021.

Natalia Silveira, Timothy Dozat, Marie-Catherine de Marneffe, Samuel Bowman, Miriam Connor, John Bauer, and Christopher D. Manning. A gold standard dependency corpus for English. In *Proceedings of the Ninth International Conference on Language Resources and Evaluation (LREC'14)*, 2014.

William Timkey and Marten van Schijndel. All bark and no bite: Rogue dimensions in transformer language models obscure representational quality. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, 2021.

Alexander Matt Turner, Lisa Thiergart, Gavin Leech, David Udell, Juan J. Vazquez, Ulisse Mini, and Monte MacDiarmid. Activation addition: Steering language models without optimization. *arXiv preprint arXiv:2308.10248*, 2023.

Elena Voita and Ivan Titov. Information-theoretic probing with minimum description length. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*, 2020.

Appendix A

Structural shuffle nulls

For paired contrasts the null direction is $\mathbf{v}_{\text{null}}^{(b)} \propto \sum_i \sigma_i^{(b)}(\mathbf{x}_i^+ - \mathbf{x}_i^-)$ with Rademacher signs $\sigma_i^{(b)}$. For unmatched data, class labels are repartitioned with group sizes preserved. Both nulls are two-sided.

Against the sign-flip null, per-concept gaps for the word-level controls ranged from about -0.05 to $+0.03$. Two of 42 word-level concepts reached $\alpha = 0.05$, consistent with the roughly two expected by chance.

Refusal (corrected). Refusal is unmatched, so it uses the label-repartition null. Across the seven models retained after the reliability gate, the mean gap was -0.0302 for the primary refusal set (t-test $p = 0.050$, Wilcoxon $p = 0.031$), -0.0397 for AdvBench ($p = 0.023$), -0.0419 for HarmBench ($p = 0.025$) and -0.0441 for JailbreakBench ($p = 0.019$). Individual model gaps ranged from -0.0995 to $+0.0052$. The four refusal sets overlap heavily in construction and are not independent replications.

Appendix B

gemma-2-2b

Against the pre-specified frozen decision rule, gemma-2-2b's refusal-direction split-half cosine was 0.06–0.44, below the threshold of 0.70, and the result did not change when the sample was doubled to 512 pairs. Its word concepts never separated at any layer (mean pairwise cosine 0.66–0.86, against 0.26 in other models). The original analysis excluded it on the concept side but kept it on the syntax side. Here every test is reported both ways (Table 5.2, `primary_and_family.csv`). In the final data this matters for one number only: the loud-tail Δ is significant with gemma-2-2b ($p = 0.025$) and not without it ($p = 0.062$). Table B.1 repeats the pre-registered family without gemma-2-2b.

Tail	Layer	Quantity	mean	95% CI	p	p_{Holm}	p_{TOST}
Loud	final	Δ (primary)	-0.034	$[-0.071, +0.002]$	0.062	0.431	0.002
Loud	final	g	$+0.024$	$[-0.014, +0.061]$	0.171	1.000	0.001
Loud	mid	Δ	-0.036	$[-0.071, -0.002]$	0.043	0.343	0.002
Loud	mid	g	$+0.004$	$[-0.022, +0.030]$	0.728	1.000	<0.001
Quiet	mid	Δ (primary)	-0.009	$[-0.084, +0.066]$	0.774	1.000	0.013
Quiet	mid	g	-0.011	$[-0.060, +0.039]$	0.608	1.000	0.002
Quiet	final	Δ	-0.030	$[-0.082, +0.022]$	0.205	1.000	0.008
Quiet	final	g	-0.010	$[-0.047, +0.027]$	0.542	1.000	<0.001

Table B.1: The pre-registered family without gemma-2-2b ($n = 7$ models).

Appendix C

Steering and rank

Activation addition. On two models, with both signs and four steering magnitudes, each real direction was compared against 50 shuffled directions generated under the corresponding null. Between 11 and 17 of the 50 shuffled directions matched the real direction; the minimum permutation p-value was 0.24. No perplexity experiment was run.

Rank. The rank hypothesis (Engels et al., 2025) gave an overall $p = 0.374$. Two models reached significance individually, in opposite directions.